



The octonion linear canonical transform: Definition and properties

Wen-Biao Gao, Bing-Zhao Li*

School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China

ARTICLE INFO

Article history:

Received 25 April 2020

Revised 8 April 2021

Accepted 29 June 2021

Available online 30 June 2021

Keywords:

Linear canonical transform
Octonion
Octonion Fourier transform
Octonion linear canonical transform
Uncertainty principle

ABSTRACT

The linear canonical transform (LCT) is a kind of integral transforms with wide applications in signal analysis. There have been numerous studies in the literature to generalize the LCT by making use of the quaternion algebra. In this paper, we first define the octonion linear canonical transform (OCLCT). Based on the definition of OCLCT, we extend the relationship between the LCT and the Fourier transform (FT) to the OCLCT and the octonion Fourier transform (OFT). Then explore related properties for the OCLCT such as shift property, inversion formula, isometry and Riemann-Lebesgue lemma. The relation between OCLCT and 3-D LCT is also build. Moreover, based on these properties, we obtain Heisenberg's uncertainty principle and Donoho-Stark's uncertainty principle associated with the OCLCT. Finally, some potential applications are presented to show the effectiveness of the OCLCT.

© 2021 Elsevier B.V. All rights reserved.

1. Introduction

Over the last few years, there has been a growing interest in hypercomplex algebras including applications in color image processing, watermarking, image filtering, edge detection and pattern recognition [1–7]. Quaternions is the Cayley-Dickson algebra of order 4 with wide application in signal processing and optical. The quaternion Fourier transform (QFT) [8] is particularly useful in quaternions algebras and known that is an alternative to the commonly used 2-D Complex Fourier Transform (CFT). Some useful properties of the QFT were established in [9]. It has found some applications in color image analysis, color image digital watermarking scheme, image pre-processing and neural computing techniques for speech recognition, etc. in [10–12]. The quaternion linear canonical transform (QLCT), as a generalization of the QFT, is another effective processing tool for signal analysis. Some useful properties of the QLCT have been established in [13–16].

Octonions, with the Cayley-Dickson algebra of order 8, is also known as Cayley-Dickson construction and is another hypercomplex algebras. Some properties of the octonion have been introduced in [17]. Recently, the octonion Fourier transform (OFT) is becoming one of the hot research topics in modern signal processing community. The OFT was first proposed in the 2011 by Hahn and Snopek [18]. As a generalization of the QFT, the OFT is defined by a triple integral with an octonion kernel function. Some interesting properties have been studied, for instance, shift property, symmetry properties, Parseval-Plancherel and Wiener-Khintchine the-

orem, uncertainty principle, etc [18–20]. The application example for the OFT has been investigated in [21,22].

To the best of our knowledge, the definition and properties of octonion linear canonical transform (OCLCT) have never been proposed up to date. In this paper, we propose the octonion linear canonical transform (OCLCT) by substituting the Fourier transform kernel with the linear canonical transform kernel. The relation between the OCLCT and the QLCT are obtained. Some important properties of the OCLCT are analyzed, for example, shift property, inversion formula, isometry, Riemann-Lebesgue lemma and the relation between OCLCT and 3-D LCT. In addition, Heisenberg's uncertainty principle and Donoho-Stark's uncertainty principle associated with the OCLCT are established. Finally, some potential applications are presented to show the effectiveness of the OCLCT.

The paper is organized as follows: In Section 2, some general definitions and basic properties of octonion algebra are summarized. The definition and the properties of the OCLCT are studied in Section 3. In Section 4, the uncertainty principles associated with the OCLCT are obtained. Some potential applications are given in Section 5. In Section 6, conclusions are drawn.

2. Preliminaries

In this section, mainly review some basic facts on the octonion algebra [23], which will be needed throughout the paper.

* Corresponding author.

E-mail address: li_bingzhao@bit.edu.cn (B.-Z. Li).

Table 1

*	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
1	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	-1	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	-1	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	-1	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	-1	e_1	e_2	e_3
e_5	e_5	e_4	$-e_7$	e_6	$-e_1$	-1	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	-1	$-e_1$
e_7	e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	-1

2.1. Octonion algebra

According to the Cayley-Dickson construction, the octonion algebra is denoted by $\mathbb{O}$ [23]. An arbitrary $o \in \mathbb{O}$ can be written in the form

$$o = o_0 + o_1 e_1 + o_2 e_2 + o_3 e_3 + o_4 e_4 + o_5 e_5 + o_6 e_6 + o_7 e_7$$

where $o_0, o_1, \dots, o_7 \in \mathbb{R}$. The octonion algebra is a non-associative and a non-commutative algebra. The multiplication rules in octonion algebra are presented in Table 1.

The octonion conjugate is defined by

$$\bar{o} = o_0 - o_1 e_1 - o_2 e_2 - o_3 e_3 - o_4 e_4 - o_5 e_5 - o_6 e_6 - o_7 e_7.$$

We can define the norm by $|o| = \sqrt{o\bar{o}} = \sqrt{\bar{o}o}$ and $|o|^2 = \sum_{r=0}^7 o_r^2$. The norm satisfies $|\epsilon\epsilon| = |\epsilon||\epsilon|$, for all $\epsilon, \epsilon \in \mathbb{O}$.

Every octonion $o \in \mathbb{O}$ can be represented in the form

$$o = \alpha + \beta e_4,$$

where $\alpha = o_0 + o_1 e_1 + o_2 e_2 + o_3 e_3 \in \mathbb{H}$ and $\beta = o_4 + o_5 e_1 + o_6 e_2 + o_7 e_3 \in \mathbb{H}$. This form is called the quaternionic form of an octonion.

The following properties are given in [17,19], they are very useful.

Property 1. Let $\alpha, \beta \in \mathbb{H}$. Then

$$(1) e_4 \alpha = \bar{\alpha} e_4; \quad (2) e_4 (\alpha) e_4 = -\bar{\alpha}; \quad (3) (\alpha e_4) e_4 = -\alpha,$$

$$(4) \alpha (\beta e_4) = (\beta \alpha) e_4; \quad (5) (\alpha e_4) \beta = (\alpha \bar{\beta}) e_4; \quad (6) (\alpha e_4) (\beta e_4) = -\bar{\beta} \alpha.$$

Property 2. For an octonion $\alpha + \beta e_4$, $\alpha, \beta \in \mathbb{H}$ in the quaternionic form we have

$$\overline{\alpha + \beta e_4} = \bar{\alpha} - \beta e_4,$$

$$|\alpha + \beta e_4|^2 = |\alpha|^2 + |\beta|^2.$$

Property 3. Let $\tilde{o}, \hat{o} \in \mathbb{O}$. Then

$$\mathbf{e}^{\tilde{o}} \cdot \mathbf{e}^{\hat{o}} = \mathbf{e}^{\tilde{o} + \hat{o}} \quad \text{if and only if} \quad \tilde{o} \cdot \hat{o} = \hat{o} \cdot \tilde{o},$$

where $\cdot$ is dot product.

An octonion valued function $f: \mathbb{R}^3 \rightarrow \mathbb{O}$ is defined by

$$f(\mathbf{x}) = f_0(\mathbf{x}) + f_1(\mathbf{x})e_1 + \dots + f_7(\mathbf{x})e_7 = \tilde{f} + \hat{f}e_4, \quad (1)$$

where each $f_z(\mathbf{x})$ is real valued functions for $z = 0, 1, \dots, 7$ and $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$.

For each octonion-valued signal $f(\mathbf{x})$ over $\mathbb{R}^3$ and $1 \leq p < \infty$, the L^p -norm of f is defined by

$$\|f\|_p^p = \int_{\mathbb{R}^3} |f(\mathbf{x})|^p d\mathbf{x},$$

if $p = \infty$, then the L^∞ -norm of f is defined by

$$\|f\|_\infty = \text{ess sup}_{\mathbf{x} \in \mathbb{R}^3} |f(\mathbf{x})|.$$

2.2. Octonion Fourier transform

The one dimensional OFT [20] of an octonion-valued function $f: \mathbb{R} \rightarrow \mathbb{O}$ is given by

$$F_1(f)(u) = \int_{\mathbb{R}} f(x) \mathbf{e}^{-e_4 2\pi x u} dx,$$

and its inverse is given by

$$f(x) = F_1^{-1}(F_1(f))(x) = \int_{\mathbb{R}} F_1(f)(u) \mathbf{e}^{e_4 2\pi x u} du. \quad (2)$$

The three dimensional OFT [20,24] of an octonion-valued function $f : \mathbb{R}^3 \rightarrow \mathbb{O}$ could be defined as

$$F_o(f)(\mathbf{u}) = \int_{\mathbb{R}^3} f(\mathbf{x}) \mathbf{e}^{-e_1 2\pi x_1 u_1} \mathbf{e}^{-e_2 2\pi x_2 u_2} \mathbf{e}^{-e_4 2\pi x_3 u_3} d\mathbf{x}, \quad (3)$$

where $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$, $\mathbf{u} = (u_1, u_2, u_3) \in \mathbb{R}^3$.

Since the octonion is non-associative and non-commutative, the order of imaginary units in the formula (3) is strictly determined and its change gives other definitions of hypercomplex FTs. The octonion 3-D signal (1) has the octonion structure, so the OFT of the octonion 3-D signal should follow the multiplication rules of octonion algebra (Table 1). Based on the octonion algebra is non-associative, it should be noted that the multiplication in the OFT is done from left to right.

Next, we present several properties of the OFT [18–21]:

- The inverse OFT of the spectrum $F_o(f)$ is

$$\begin{aligned} f(\mathbf{x}) &= F_o^{-1}(F_o(f))(\mathbf{x}) \\ &= \int_{\mathbb{R}^3} F_o(f)(\mathbf{u}) \mathbf{e}^{e_4 2\pi x_3 u_3} \mathbf{e}^{e_2 2\pi x_2 u_2} \mathbf{e}^{e_1 2\pi x_1 u_1} d\mathbf{u}, \end{aligned} \quad (4)$$

where $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$, $\mathbf{u} = (u_1, u_2, u_3) \in \mathbb{R}^3$.

- The Parseval theorem for OFT is

$$\|f\|_2 = \|F_o f\|_2.$$

- For $1 \leq p \leq 2$ and $\frac{1}{p} + \frac{1}{q} = 1$, the Hausdorff-Young inequality for OFT is

$$\|F_o f\|_q \leq \|f\|_p.$$

- The relation between the OFT and the 3-D FT is that

$$\begin{aligned} F_o(f)(u_1, u_2, u_3) &= \frac{1}{4} \left[F(u_1, u_2, u_3) + F(u_1, -u_2, u_3) \right] (1 - e_5) \\ &\quad + \frac{1}{4} \left[F(u_1, u_2, -u_3) + F(u_1, -u_2, -u_3) \right] (1 + e_5) \\ &\quad + \frac{1}{4} e_3 \left[F(-u_1, u_2, u_3) - F(-u_1, -u_2, u_3) \right] (1 + e_5) \\ &\quad + \frac{1}{4} e_3 \left[F(-u_1, u_2, -u_3) - F(-u_1, -u_2, -u_3) \right] (1 - e_5), \end{aligned}$$

where $F(u_1, u_2, u_3) = \int_{\mathbb{R}^3} f(\mathbf{x}) \mathbf{e}^{-e_1 2\pi x_1 u_1} \mathbf{e}^{-e_1 2\pi x_2 u_2} \mathbf{e}^{-e_1 2\pi x_3 u_3} d\mathbf{x}$ is the 3-D FT. It shows that the level of computational complexity of OFT calculation remains almost the same as in the classical Fourier transform. The formulas relating complex and hypercomplex FTs are very important from a practical point of view. They can be used to derive quaternion and octonion Fourier spectra starting with 2-D and 3-D complex FTs [25].

The theories of OFT can be generalized to the case of functions with hypercomplex values. The OFT has symmetry properties expressed in terms of octonion anti-involutions. It implies that the whole information about a 3-D signal is limited to a single octant of the frequency space [1,18,25]. Furthermore, some properties of the OFT are used in the analysis of some three-dimensional linear time-invariant systems described by systems of partial differential equations with constant coefficients [21].

2.3. Linear canonical transform

Let e_1, e_2 and e_3 (or equivalently i, j, k) denote the three imaginary units in the quaternion algebra [26]. We first show that the linear canonical transform (LCT) of a function $f : \mathbb{R} \rightarrow \mathbb{O}$

$$\mathcal{L}_A\{f\}(u) = \int_{\mathbb{R}} f(x) K_A^{e_1}(x, u) dx,$$

and its inverse is given by

$$\begin{aligned} f(x) &= \mathcal{L}_A^{-1}[\mathcal{L}_A\{f\}](x) \\ &= \int_{\mathbb{R}} \mathcal{L}_A\{f\}(u) K_{A^{-1}}^{e_1}(u, x) du, \end{aligned}$$

where $x, u \in \mathbb{R}$, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A) = 1$, $A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ and the kernel signal

$$K_A^{e_1}(x, u) := \begin{cases} \frac{1}{\sqrt{2\pi|b|}} \mathbf{e}^{e_1 \left(\frac{a}{2b} x^2 - \frac{xu}{b} + \frac{d}{2b} u^2 - \frac{\pi}{4} \right)}, & b \neq 0 \\ \sqrt{d} \mathbf{e}^{e_1 \frac{cd}{2} u^2} \delta(x - du), & b = 0 \end{cases},$$

for $\delta(x)$ representing the Dirac function.

The LCT can be generalized to QLCT, the QLCT of a function $f : \mathbb{R}^2 \rightarrow \mathbb{O}$ is defined by

$$\mathcal{L}_{A_1, A_2}\{f\}(\mathbf{u}) = \int_{\mathbb{R}^2} f(\mathbf{x}) K_{L, A_1}^{e_1}(x_1, u_1) K_{L, A_2}^{e_2}(x_2, u_2) d\mathbf{x}, \quad (5)$$

and its inverse is given by

$$f(\mathbf{x}) = \mathcal{L}_{A_1, A_2}^{-1}[\mathcal{L}_{A_1, A_2}\{f\}](\mathbf{x}) \quad (6)$$

$$= \int_{\mathbb{R}^2} \mathcal{L}_{A_1, A_2} \{f\}(\mathbf{u}) K_{L, A_2^{-1}}^{e_2}(u_2, x_2) K_{L, A_1^{-1}}^{e_1}(u_1, x_1) d\mathbf{u}, \quad (6)$$

where $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$, $\mathbf{u} = (u_1, u_2) \in \mathbb{R}^2$, $A_s = \begin{bmatrix} a_s & b_s \\ c_s & d_s \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_s) = 1$, $A_s^{-1} = \begin{bmatrix} d_s & -b_s \\ -c_s & a_s \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ and the kernel signal

$$K_{L, A_s}^{e_s}(x_s, u_s) := \begin{cases} \frac{1}{\sqrt{2\pi|b_s|}} e^{e_s \left(\frac{a_s}{2b_s} x_s^2 - \frac{x_s u_s}{b_s} + \frac{d_s}{2b_s} u_s^2 - \frac{\pi}{4} \right)}, & b_s \neq 0 \\ \sqrt{d_s} e^{e_s \frac{c_s d_s}{2} u_s^2} \delta(x_s - d_s u_s), & b_s = 0 \end{cases}, \quad (7)$$

for $s = 1, 2$, $\delta(x)$ representing the Dirac function.

However, there has been no work studying the LCT in the octonion algebra analysis setting. Next, we will research the OCLCT.

3. Octonion linear canonical transform

3.1. The definition of OCLCT

According to the one dimensional OFT, we can obtain the definition of one dimensional OCLCT as follows:

Definition 1. Let an octonion-valued function $f: \mathbb{R} \rightarrow \mathbb{O}$, then

$$\mathcal{L}_{1, A}^o(f)(u) = \int_{\mathbb{R}} f(x) K_A^{e_4}(x, u) dx, \quad (8)$$

where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A) = 1$, and the kernel signal

$$K_A^{e_4}(x, u) := \begin{cases} \frac{1}{\sqrt{2\pi|b|}} e^{e_4 \left(\frac{a}{2b} x^2 - \frac{xu}{b} + \frac{d}{2b} u^2 - \frac{\pi}{2} \right)}, & b \neq 0 \\ \sqrt{d} e^{e_4 \frac{cd}{2} u^2} \delta(x - du), & b = 0 \end{cases}. \quad (9)$$

In the following we obtain the relationship between the one dimensional OFT and the one dimensional OCLCT of an octonion-valued signal $f(x)$.

$$\mathcal{L}_{1, A}^o(f)(u) = \frac{1}{\sqrt{2\pi|b|}} F_1(g) \left(\frac{u}{2\pi|b|} \right) e^{e_4 \left(\frac{d}{2b} u^2 - \frac{\pi}{4} \right)}, \quad (10)$$

where $b \neq 0$ and $g(x) = f(x) e^{e_4 \frac{a}{2b} x^2}$.

Theorem 1 (Inversion formula of the one dimensional OCLCT). *Let an octonion valued function $f: \mathbb{R} \rightarrow \mathbb{O}$. Then we have the inversion formula of the one dimensional OCLCT,*

$$f(x) = \int_{\mathbb{R}} \mathcal{L}_{1, A}^o(f)(u) K_A^{-e_4}(x, u) du. \quad (11)$$

where $K_A^{-e_4}(x, u) = K_{A^{-1}}^{e_4}(u, x) = \frac{1}{\sqrt{2\pi|b|}} e^{-e_4 \left(\frac{a}{2b} x^2 - \frac{xu}{b} + \frac{d}{2b} u^2 - \frac{\pi}{2} \right)}$,

$$A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \in \mathbb{R}^{2 \times 2} \text{ and } b \neq 0.$$

Proof. According to (10), we have

$$\sqrt{2\pi|b|} \mathcal{L}_{1, A}^o(f)(u) e^{-e_4 \left(\frac{d}{2b} u^2 - \frac{\pi}{2} \right)} = F_1(g) \left(\frac{u}{2\pi|b|} \right),$$

where $b \neq 0$ and $g(x) = f(x) e^{e_4 \frac{a}{2b} x^2}$.

Using (2), it follows that

$$f(x) e^{e_4 \frac{a}{2b} x^2} = \int_{\mathbb{R}} \sqrt{2\pi|b|} \mathcal{L}_{1, A}^o(f)(u) e^{-e_4 \left(\frac{d}{2b} u^2 - \frac{\pi}{2} \right)} e^{e_4 x \frac{u}{b}} d \left(\frac{u}{2\pi|b|} \right).$$

Hence

$$f(x) = \int_{\mathbb{R}} \mathcal{L}_{1, A}^o(f)(u) \frac{1}{\sqrt{2\pi|b|}} e^{-e_4 \left(\frac{d}{2b} u^2 - \frac{\pi}{2} \right)} e^{e_4 2\pi x \frac{u}{2\pi b}} e^{-e_4 \frac{a}{2b} x^2} du,$$

which completes the proof. $\square$

Next, we will give the three dimensional OCLCT.

Definition 2. Let an octonion-valued signal $f: \mathbb{R}^3 \rightarrow \mathbb{O}$, then

$$\mathcal{L}_{A_1, A_2, A_3}^o\{f\}(\mathbf{u}) = \int_{\mathbb{R}^3} f(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) K_{A_3}^{e_4}(x_3, u_3) d\mathbf{x}, \quad (12)$$

where $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$, $\mathbf{u} = (u_1, u_2, u_3) \in \mathbb{R}^3$, $A_k = \begin{bmatrix} a_k & b_k \\ c_k & d_k \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_k) = 1$, for $k = 1, 2, 3$ and the kernel signals

$$K_{A_1}^{e_1}(x_1, u_1) := \begin{cases} \frac{1}{\sqrt{2\pi|b_1|}} \mathbf{e}^{e_1 \left(\frac{a_1}{2b_1} x_1^2 - \frac{x_1 u_1}{b_1} + \frac{d_1}{2b_1} u_1^2 - \frac{\pi}{2} \right)}, & b_1 \neq 0 \\ \sqrt{d_1} \mathbf{e}^{e_1 \frac{c_1 d_1}{2} u_1^2} \delta(x_1 - d_1 u_1), & b_1 = 0 \end{cases}, \quad (13)$$

$$K_{A_2}^{e_2}(x_2, u_2) := \begin{cases} \frac{1}{\sqrt{2\pi|b_2|}} \mathbf{e}^{e_2 \left(\frac{a_2}{2b_2} x_2^2 - \frac{x_2 u_2}{b_2} + \frac{d_2}{2b_2} u_2^2 - \frac{\pi}{2} \right)}, & b_2 \neq 0 \\ \sqrt{d_2} \mathbf{e}^{e_2 \frac{c_2 d_2}{2} u_2^2} \delta(x_2 - d_2 u_2), & b_2 = 0 \end{cases}, \quad (14)$$

and

$$K_{A_3}^{e_4}(x_3, u_3) := \begin{cases} \frac{1}{\sqrt{2\pi|b_3|}} \mathbf{e}^{e_4 \left(\frac{a_3}{2b_3} x_3^2 - \frac{x_3 u_3}{b_3} + \frac{d_3}{2b_3} u_3^2 - \frac{\pi}{2} \right)}, & b_3 \neq 0 \\ \sqrt{d_3} \mathbf{e}^{e_4 \frac{c_3 d_3}{2} u_3^2} \delta(x_3 - d_3 u_3), & b_3 = 0 \end{cases}, \quad (15)$$

where $\delta(x)$ representing the Dirac function.

Note that, for $b_k = 0 (k = 1, 2, 3)$, the OCLCT of a signal is essentially a chirp multiplication and it is of no particular interest for our objective in this work. Hence, without loss of generality, we set $b_k \neq 0$ in the following sections unless stated otherwise.

Theorem 2 (Inversion formula of the three dimensional OCLCT). *Let an octonion valued function $f : \mathbb{R}^3 \rightarrow \mathbb{O}$. Then we have the inversion formula of the three dimensional OCLCT,*

$$\begin{aligned} f(\mathbf{x}) &= (\mathcal{L}_{A_1, A_2, A_3}^o)^{-1} [\mathcal{L}_{A_1, A_2, A_3}^o \{f\}](\mathbf{x}) \\ &= \int_{\mathbb{R}^3} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{u}) K_{A_3}^{e_4}(u_3, x_3) K_{A_2}^{e_2}(u_2, x_2) K_{A_1}^{e_1}(u_1, x_1) d\mathbf{u}, \end{aligned} \quad (16)$$

where $A_k^{-1} = \begin{bmatrix} d_k & -b_k \\ -c_k & a_k \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, for $k = 1, 2, 3$.

Proof. From the definitions of the QLCT, the one dimensional OCLCT and the three dimensional OCLCT, we have

$$\begin{aligned} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{u}) &= \int_{\mathbb{R}} \mathcal{L}_{A_1, A_2} \{f\}(u_1, u_2) K_{A_3}^{e_4}(x_3, u_3) dx_3 \\ &= \mathcal{L}_{1, A_3}^o [\mathcal{L}_{A_1, A_2} \{f\}](\mathbf{u}). \end{aligned} \quad (17)$$

By assumption $f : \mathbb{R}^3 \rightarrow \mathbb{O}$, then $\mathcal{L}_{A_1, A_2} \{f\} : \mathbb{R}^3 \rightarrow \mathbb{O}$. Therefore,

$$\begin{aligned} &\int_{\mathbb{R}^3} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{u}) K_{A_3}^{e_4}(u_3, x_3) K_{A_2}^{e_2}(u_2, x_2) K_{A_1}^{e_1}(u_1, x_1) d\mathbf{u} \\ &= \int_{\mathbb{R}^2} \int_{\mathbb{R}} \mathcal{L}_{1, A_3}^o [\mathcal{L}_{A_1, A_2} \{f\}](\mathbf{u}) K_{A_3}^{e_4}(u_3, x_3) du_3 \\ &\times K_{A_2}^{e_2}(u_2, x_2) K_{A_1}^{e_1}(u_1, x_1) du_1 du_2. \end{aligned}$$

According to Theorem 1 and (6), we obtain

$$\begin{aligned} &\int_{\mathbb{R}^3} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{u}) K_{A_3}^{e_4}(u_3, x_3) K_{A_2}^{e_2}(u_2, x_2) K_{A_1}^{e_1}(u_1, x_1) d\mathbf{u} \\ &= \int_{\mathbb{R}^2} \mathcal{L}_{A_1, A_2} \{f\}(u_1, u_2, x_3) K_{A_2}^{e_2}(u_2, x_2) K_{A_1}^{e_1}(u_1, x_1) du_1 du_2 \\ &= f(\mathbf{x}), \end{aligned}$$

which completes the proof. $\square$

Remark 1. When $a_j = d_j = 0, b_j = 1, c_j = -1$, for $j = 1, 2, 3$, the OCLCT reduces to the OFT. That is

$$\mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{u}) = \frac{1}{2\pi\sqrt{2\pi}} F_0(f) \left(\frac{u_1}{2\pi}, -\frac{u_2}{2\pi}, -\frac{u_3}{2\pi} \right) e_7.$$

3.2. Properties of the OCLCT

We can expand the kernel of the OCLCT in the form

$$\begin{aligned} &(K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2)) K_{A_3}^{e_4}(x_3, u_3) \\ &= \frac{1}{2\pi\sqrt{2\pi|b_1 b_2 b_3|}} (\mathbf{e}^{e_1 \gamma_1} \mathbf{e}^{e_2 \gamma_2}) \mathbf{e}^{e_4 \gamma_3} \\ &= \frac{1}{2\pi\sqrt{2\pi|b_1 b_2 b_3|}} ((\cos \gamma_1 + e_1 \sin \gamma_1)(\cos \gamma_2 \end{aligned}$$

$$\begin{aligned}
& + e_2 \sin \gamma_2))(\cos \gamma_3 + e_4 \sin \gamma_3) \\
& = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} (\cos \gamma_1 \cos \gamma_2 \cos \gamma_3 \\
& + \sin \gamma_1 \cos \gamma_2 \cos \gamma_3 e_1 \\
& + \cos \gamma_1 \sin \gamma_2 \cos \gamma_3 e_2 \\
& + \sin \gamma_1 \sin \gamma_2 \cos \gamma_3 e_3 \\
& + \cos \gamma_1 \cos \gamma_2 \sin \gamma_3 e_4 \\
& + \sin \gamma_1 \cos \gamma_2 \sin \gamma_3 e_5 \\
& + \cos \gamma_1 \sin \gamma_2 \sin \gamma_3 e_6 \\
& + \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 e_7), \tag{18}
\end{aligned}$$

where $\gamma_k = \frac{a_k}{2b_k} x_k^2 - \frac{x_k u_k}{b_k} + \frac{d_k}{2b_k} u_k^2 - \frac{\pi}{2}$, $k = 1, 2, 3$. The form (18) is a full octonion as defined in (12). It is worth noticing that in the definition of the OCLCT one can use any of the 28 octonion triples that doesn't form an associative, quaternion, non-commutative subfield of $\mathbb{O}$.

The OCLCT of an octonion valued function $f: \mathbb{R}^3 \rightarrow \mathbb{O}$ of three variables can be expressed, as follows

$$\begin{aligned}
L &= \mathcal{L}_{A_1, A_2, A_3}^o \{f\} \\
&= L_{eee} + L_{oeo} e_1 + L_{eoe} e_2 + L_{ooo} e_3 + L_{eeo} e_4 \\
&+ L_{oeo} e_5 + L_{eoo} e_6 + L_{ooo} e_7, \tag{19}
\end{aligned}$$

where

$$L_{eee}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{eee}(\mathbf{x}) \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x},$$

$$L_{oeo}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{oeo}(\mathbf{x}) \sin \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x},$$

$$L_{eoe}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{eoe}(\mathbf{x}) \cos \gamma_1 \sin \gamma_2 \cos \gamma_3 d\mathbf{x},$$

$$L_{ooo}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{ooo}(\mathbf{x}) \sin \gamma_1 \sin \gamma_2 \cos \gamma_3 d\mathbf{x},$$

$$L_{eeo}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{eeo}(\mathbf{x}) \cos \gamma_1 \cos \gamma_2 \sin \gamma_3 d\mathbf{x},$$

$$L_{eoo}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{eoo}(\mathbf{x}) \sin \gamma_1 \cos \gamma_2 \sin \gamma_3 d\mathbf{x},$$

$$L_{ooo}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{ooo}(\mathbf{x}) \cos \gamma_1 \sin \gamma_2 \sin \gamma_3 d\mathbf{x},$$

$$L_{ooo}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f_{ooo}(\mathbf{x}) \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 d\mathbf{x}.$$

We use subscripts e and o to indicate that a function is either even (e) or odd (o) with respect to an appropriate variable, i.e. $f_{oeo}(\mathbf{x})$ is odd with respect to x_1 and x_2 and even with respect to x_3 .

Consider the OCLCT of the function shifted with respect to the first variable, i.e. $f^{r_1}(x_1, x_2, x_3) = f(x_1 - r_1, x_2, x_3)$. The shift with respect to the first variable does not change the parity with respect to other remaining variables. We will prove the shift theorem of the OCLCT.

Theorem 3 (Shift). *Let L be the OCLCT of an octonion valued function f of three variable. Let L^{r_1} , L^{r_2} and L^{r_3} denote the OCLCT of $f(x_1 - r_1, x_2, x_3)$, $f(x_1, x_2 - r_2, x_3)$ and $f(x_1, x_2, x_3 - r_3)$, respectively. Then*

$$L^{r_1}(\mathbf{u}) = \cos \phi_1 L(\mathbf{m}) - \sin \phi_1 \Delta_1(\mathbf{m}),$$

$$L^{r_2}(\mathbf{u}) = \cos \phi_2 L(\mathbf{n}) - \sin \phi_2 \Delta_2(\mathbf{n}),$$

$$L^{r_3}(\mathbf{u}) = \cos \phi_3 L(\mathbf{p}) - \sin \phi_3 \Delta_3(\mathbf{p}),$$

where $\mathbf{m} = (m_1, u_2, u_3)$, $\mathbf{n} = (u_1, m_2, u_3)$, $\mathbf{p} = (u_1, u_2, m_3)$, $m_k = u_k - r_k a_k (k = 1, 2, 3)$ and $\Delta_1 = L_{see} - L_{cee}e_1 + L_{soe}e_2 - L_{coe}e_3 + L_{seo}e_4 - L_{ceo}e_5 + L_{sso}e_6 - L_{cso}e_7$, $\Delta_2 = L_{ese} + L_{ose}e_1 - L_{ece}e_2 - L_{oce}e_3 + L_{eso}e_4 + L_{oso}e_5 - L_{eco}e_6 - L_{oco}e_7$, $\Delta_3 = L_{ees} + L_{oes}e_1 + L_{eos}e_2 + L_{oos}e_3 - L_{eec}e_4 - L_{oec}e_5 - L_{eoc}e_6 - L_{ooc}e_7$.

Proof. We express the OCLCT of the function f^{r_1} as in (19). So

$$\begin{aligned} L_{eee}^{r_1}(\mathbf{u}) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{eee}^{r_1}(x_1, x_2, x_3) \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x} \\ &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{eee}(x_1 - r_1, x_2, x_3) \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x}, \end{aligned}$$

where $\gamma_k = \frac{a_k}{2b_k}x_k^2 - \frac{x_k u_k}{b_k} + \frac{d_k}{2b_k}u_k^2 - \frac{\pi}{2}$, $k = 1, 2, 3$. Let us consider the above integral, by making the change of variable $t_1 = x_1 - r_1$, $t_2 = x_2$, $t_3 = x_3$ ($\mathbf{t} = (t_1, t_2, t_3) \in \mathbb{R}^3$), we obtain

$$\begin{aligned} L_{eee}^{r_1}(\mathbf{u}) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{eee}(t_1, t_2, t_3) \cos(\phi_1 + \varphi_1) \cos \psi_2 \cos \psi_3 d\mathbf{t} \\ &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \left[\cos \phi_1 \int_{\mathbb{R}^3} f_{eee}(t_1, t_2, t_3) \cos \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{t} \right. \\ &\quad \left. - \sin \phi_1 \int_{\mathbb{R}^3} f_{eee}(t_1, t_2, t_3) \sin \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{t} \right], \end{aligned}$$

where $\phi_1 = r_1 u_1 c_1 - \frac{a_1 c_1 r_1^2}{2}$, $\varphi_1 = \frac{a_1}{2b_1}t_1^2 - \frac{t_1(u_1 - r_1 a_1)}{b_1} + \frac{d_1}{2b_1}(u_1 - r_1 a_1)^2 - \frac{\pi}{2}$, $\psi_k = \frac{a_k}{2b_k}x_k^2 - \frac{x_k u_k}{b_k} + \frac{d_k}{2b_k}u_k^2 - \frac{\pi}{2}$, $k = 2, 3$. Let

$$\begin{aligned} L_{see}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{eee}(\mathbf{t}) \sin \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{t}, \\ L_{cee}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{cee}(\mathbf{t}) \cos \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{t}, \\ L_{soe}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{soe}(\mathbf{t}) \sin \varphi_1 \sin \psi_2 \cos \psi_3 d\mathbf{t}, \\ L_{coe}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{coe}(\mathbf{t}) \cos \varphi_1 \sin \psi_2 \cos \psi_3 d\mathbf{t}, \\ L_{seo}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{seo}(\mathbf{t}) \sin \varphi_1 \cos \psi_2 \sin \psi_3 d\mathbf{t}, \\ L_{ceo}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{ceo}(\mathbf{t}) \cos \varphi_1 \cos \psi_2 \sin \psi_3 d\mathbf{t}, \\ L_{sso}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{sso}(\mathbf{t}) \sin \varphi_1 \sin \psi_2 \sin \psi_3 d\mathbf{t}, \\ L_{cso}(m_1, u_2, u_3) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{cso}(\mathbf{t}) \cos \varphi_1 \sin \psi_2 \sin \psi_3 d\mathbf{t}. \end{aligned}$$

Then

$$L_{eee}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{eee}(m_1, u_2, u_3) - \sin \phi_1 L_{see}(m_1, u_2, u_3),$$

where $m_1 = u_1 - r_1 a_1$.

Similarly we can prove that

$$\begin{aligned} L_{oee}^{r_1}(\mathbf{u}) &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} f_{oee}(t_1, t_2, t_3) \sin(\phi_1 + \varphi_1) \cos \psi_2 \cos \psi_3 d\mathbf{t} \\ &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \left[\sin \phi_1 \int_{\mathbb{R}^3} f_{oee}(t_1, t_2, t_3) \cos \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{t} \right. \\ &\quad \left. + \cos \phi_1 \int_{\mathbb{R}^3} f_{oee}(t_1, t_2, t_3) \sin \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{t} \right] \\ &= \cos \phi_1 L_{oee}(m_1, u_2, u_3) + \sin \phi_1 L_{cee}(m_1, u_2, u_3). \end{aligned}$$

Summarizing, we have

$$L_{eee}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{eee}(m_1, u_2, u_3) - \sin \phi_1 L_{see}(m_1, u_2, u_3),$$

$$L_{oe}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{oe}(m_1, u_2, u_3) + \sin \phi_1 L_{cee}(m_1, u_2, u_3),$$

$$L_{eo}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{eo}(m_1, u_2, u_3) - \sin \phi_1 L_{soe}(m_1, u_2, u_3),$$

$$L_{oe}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{oe}(m_1, u_2, u_3) + \sin \phi_1 L_{coe}(m_1, u_2, u_3),$$

$$L_{eo}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{eo}(m_1, u_2, u_3) - \sin \phi_1 L_{seo}(m_1, u_2, u_3),$$

$$L_{oe}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{oe}(m_1, u_2, u_3) + \sin \phi_1 L_{ceo}(m_1, u_2, u_3),$$

$$L_{eo}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{eo}(m_1, u_2, u_3) - \sin \phi_1 L_{soo}(m_1, u_2, u_3),$$

$$L_{oe}^{r_1}(\mathbf{u}) = \cos \phi_1 L_{oe}(m_1, u_2, u_3) + \sin \phi_1 L_{coo}(m_1, u_2, u_3).$$

By (19), we obtain

$$L^{r_1}(\mathbf{u}) = \cos \phi_1 L(\mathbf{m}) - \sin \phi_1 \Delta_1(\mathbf{m}),$$

where $\mathbf{m} = (m_1, u_2, u_3)$ and $\Delta_1 = L_{see} - L_{cee}e_1 + L_{soe}e_2 - L_{coe}e_3 + L_{seo}e_4 - L_{ceo}e_5 + L_{soo}e_6 - L_{coo}e_7$.

Analogously method hold for concerning shifts with respect to variables x_2 and x_3 .

Which completes the proof. $\square$

Denoting the even and odd part of a function $f(\mathbf{x})$ by $f_e(\mathbf{x})$ and $f_o(\mathbf{x})$, where $f_e(\mathbf{x}) = \frac{f(x_1, x_2, x_3) + f(x_1, x_2, -x_3)}{2}$ is only even in the third variable x_3 and $f_o(\mathbf{x}) = \frac{f(x_1, x_2, x_3) - f(x_1, x_2, -x_3)}{2}$ is only odd in the third variable x_3 . The octonion valued function f can be represented in the quaternion form $f(\mathbf{x}) = \tilde{f} + \hat{f}e_4$. From the Lemma 2.1, the OCLCT of f becomes,

$$\begin{aligned} \mathcal{L}_{A_1, A_2, A_3}^o\{f\}(\mathbf{u}) &= \int_{\mathbb{R}^3} (\tilde{f} + \hat{f}e_4)(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) K_{A_3}^{e_4}(x_3, u_3) d\mathbf{x} \\ &= \int_{\mathbb{R}^3} \tilde{f}(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) K_{A_3}^{e_4}(x_3, u_3) d\mathbf{x} \\ &\quad + \int_{\mathbb{R}^3} \hat{f}(\mathbf{x}) K_{A_1}^{-e_1}(x_1, u_1) K_{A_2}^{-e_2}(x_2, u_2) e_4 K_{A_3}^{e_4}(x_3, u_3) d\mathbf{x}. \end{aligned} \quad (20)$$

By the even and odd part, the above formula becomes,

$$\begin{aligned} \mathcal{L}_{A_1, A_2, A_3}^o\{f\}(\mathbf{u}) &= \int_{\mathbb{R}^3} \tilde{f}_e(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \frac{1}{\sqrt{2\pi|b_3|}} \cos \gamma_3 d\mathbf{x} \\ &\quad + \int_{\mathbb{R}^3} \hat{f}_o(\mathbf{x}) K_{A_1}^{-e_1}(x_1, u_1) K_{A_2}^{-e_2}(x_2, u_2) \frac{1}{\sqrt{2\pi|b_3|}} \sin \gamma_3 d\mathbf{x} \\ &\quad + \left(\int_{\mathbb{R}^3} \tilde{f}_e(\mathbf{x}) K_{A_1}^{-e_1}(x_1, u_1) K_{A_2}^{-e_2}(x_2, u_2) \frac{1}{\sqrt{2\pi|b_3|}} \cos \gamma_3 d\mathbf{x} \right. \\ &\quad \left. - \int_{\mathbb{R}^3} \tilde{f}_o(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \frac{1}{\sqrt{2\pi|b_3|}} \sin \gamma_3 d\mathbf{x} \right) e_4, \end{aligned} \quad (21)$$

where $\gamma_3 = \frac{a_3}{2b_3}x_3^2 - \frac{x_3u_3}{b_3} + \frac{d_3}{2b_3}u_3^2 - \frac{\pi}{2}$. Hence, the OCLCT can be divided into four QLCTs, we have

$$\begin{aligned} |\mathcal{L}_{A_1, A_2, A_3}^o\{f\}(\mathbf{u})|^2 &= \frac{1}{2\pi|b_3|} \left(|\mathcal{L}_{A_1, A_2}^o\{\tilde{f}_e\}(\mathbf{u})|^2 + |\mathcal{L}_{A_1, A_2}^o\{\hat{f}_o\}(\mathbf{u})|^2 \right. \\ &\quad \left. + |\mathcal{L}_{A_1, A_2}^o\{\tilde{f}_e\}(\mathbf{u})|^2 + |\mathcal{L}_{A_1, A_2}^o\{\tilde{f}_o\}(\mathbf{u})|^2 \right). \end{aligned} \quad (22)$$

Theorem 4. Let an octonion valued function $f: \mathbb{R}^3 \rightarrow \mathbb{O}$ be a continuous and square integrable function. Then we have

$$\|f\|_2^2 = 2\pi|b_3| \|\mathcal{L}_{A_1, A_2, A_3}^o\{f\}\|_2^2. \quad (23)$$

Proof. According to Lemma 2.2 and (21), the norm of OCLCT splits into four norms of quaternion functions.

$$\begin{aligned} \|\mathcal{L}_{A_1, A_2, A_3}^o\{f\}\|_2^2 &= \frac{1}{2\pi|b_3|} \left\| \int_{\mathbb{R}^3} \tilde{f}_e(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \cos \gamma_3 d\mathbf{x} \right. \\ &\quad \left. + \int_{\mathbb{R}^3} \hat{f}_o(\mathbf{x}) K_{A_1}^{-e_1}(x_1, u_1) K_{A_2}^{-e_2}(x_2, u_2) \sin \gamma_3 d\mathbf{x} \right\|_2^2 \\ &\quad + \frac{1}{2\pi|b_3|} \left\| \int_{\mathbb{R}^3} \tilde{f}_e(\mathbf{x}) K_{A_1}^{-e_1}(x_1, u_1) K_{A_2}^{-e_2}(x_2, u_2) \cos \gamma_3 d\mathbf{x} \right. \\ &\quad \left. - \int_{\mathbb{R}^3} \tilde{f}_o(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \sin \gamma_3 d\mathbf{x} \right\|_2^2 \end{aligned}$$

$$- \int_{\mathbb{R}^3} \tilde{f}_o(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \sin \gamma_3 \mathbf{d}\mathbf{x} \Big\|_2^2. \quad (24)$$

Due to f_e and f_o are orthogonal in L^2 inner product, it follows that

$$\begin{aligned} \|\mathcal{L}_{A_1, A_2, A_3}^o \{f\}\|_2^2 &= \frac{1}{2\pi |b_3|} \left\| \int_{\mathbb{R}^3} \tilde{f}_e(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \cos \gamma_3 \mathbf{d}\mathbf{x} \right\|_2^2 \\ &\quad + \frac{1}{2\pi |b_3|} \left\| \int_{\mathbb{R}^3} \hat{f}_o(\mathbf{x}) K_{A_1}^{-e_1}(x_1, u_1) K_{A_2}^{-e_2}(x_2, u_2) \sin \gamma_3 \mathbf{d}\mathbf{x} \right\|_2^2 \\ &\quad + \frac{1}{2\pi |b_3|} \left\| \int_{\mathbb{R}^3} \hat{f}_e(\mathbf{x}) K_{A_1}^{-e_1}(x_1, u_1) K_{A_2}^{-e_2}(x_2, u_2) \cos \gamma_3 \mathbf{d}\mathbf{x} \right\|_2^2 \\ &\quad + \frac{1}{2\pi |b_3|} \left\| \int_{\mathbb{R}^3} \tilde{f}_o(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \sin \gamma_3 \mathbf{d}\mathbf{x} \right\|_2^2 \\ &= \frac{1}{2\pi |b_3|} \left(\|\mathcal{L}_{A_1, A_2}^o \{\tilde{f}_e\}\|_2^2 + \|\mathcal{L}_{A_1, A_2}^o \{\hat{f}_o\}\|_2^2 \right. \\ &\quad \left. + \|\mathcal{L}_{A_1, A_2}^o \{\hat{f}_e\}\|_2^2 + \|\mathcal{L}_{A_1, A_2}^o \{\tilde{f}_o\}\|_2^2 \right). \end{aligned} \quad (25)$$

Using the Plancherel theorem for the QLCT [27], we have

$$\left\| \int_{\mathbb{R}^3} \tilde{f}_e(\mathbf{x}) K_{A_1}^{e_1}(x_1, u_1) K_{A_2}^{e_2}(x_2, u_2) \cos \gamma_3 \mathbf{d}\mathbf{x} \right\|_2^2 = \|\tilde{f}_e\|_2^2. \quad (26)$$

The proof of the remaining three norms will be omitted, being similar to the previous one.

Furthermore, we have

$$\begin{aligned} \|f\|_2^2 &= \|\tilde{f} + \hat{f}e_4\|_2^2 \\ &= \|\tilde{f}\|_2^2 + \|\hat{f}\|_2^2 \\ &= \|\tilde{f}_e\|_2^2 + \|\tilde{f}_o\|_2^2 + \|\hat{f}_e\|_2^2 + \|\hat{f}_o\|_2^2. \end{aligned} \quad (27)$$

Combing (25), (26) and (27), we then derive the following result

$$\|f\|_2^2 = 2\pi |b_3| \|\mathcal{L}_{A_1, A_2, A_3}^o \{f\}\|_2^2.$$

Which completes the proof. $\square$

In the following, we obtain the Riemann-Lebesgue lemma of the OCLCT.

Theorem 5. (The Riemann-Lebesgue lemma of the OCLCT) Let an octonion valued function $f: \mathbb{R}^3 \rightarrow \mathbb{O}$. Then $\mathcal{L}_{A_1, A_2, A_3}^o \{f\}$ satisfies

$$\lim_{|u_1| \rightarrow \infty} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(u_1, u_2, u_3) = 0, \quad (28)$$

$$\lim_{|u_2| \rightarrow \infty} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(u_1, u_2, u_3) = 0, \quad (29)$$

and

$$\lim_{|u_3| \rightarrow \infty} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(u_1, u_2, u_3) = 0. \quad (30)$$

Proof. From the OCLCT can be divided into four QLCTs in (22), we have

$$\begin{aligned} &\lim_{|u_1| \rightarrow \infty} |\mathcal{L}_{A_1, A_2, A_3}^o \{f\}(u_1, u_2, u_3)|^2 \\ &= \frac{1}{2\pi |b_3|} \lim_{|u_1| \rightarrow \infty} \left(|\mathcal{L}_{A_1, A_2}^o \{\tilde{f}_e\}(u_1, u_2, u_3)|^2 + |\mathcal{L}_{A_1, A_2}^o \{\hat{f}_o\}(u_1, u_2, u_3)|^2 \right. \\ &\quad \left. + |\mathcal{L}_{A_1, A_2}^o \{\hat{f}_e\}(u_1, u_2, u_3)|^2 + |\mathcal{L}_{A_1, A_2}^o \{\tilde{f}_o\}(u_1, u_2, u_3)|^2 \right). \end{aligned} \quad (31)$$

Using the Riemann-Lebesgue lemma of the QLCT in [28], we obtain

$$\begin{aligned} &\lim_{|u_1| \rightarrow \infty} |\mathcal{L}_{A_1, A_2}^o \{\tilde{f}_e\}(u_1, u_2, u_3)| \\ &= \lim_{|u_1| \rightarrow \infty} |\mathcal{L}_{A_1, A_2}^o \{\hat{f}_o\}(u_1, u_2, u_3)| \\ &= \lim_{|u_1| \rightarrow \infty} |\mathcal{L}_{A_1, A_2}^o \{\hat{f}_e\}(u_1, u_2, u_3)| \\ &= \lim_{|u_1| \rightarrow \infty} |\mathcal{L}_{A_1, A_2}^o \{\tilde{f}_o\}(u_1, u_2, u_3)| = 0. \end{aligned} \quad (32)$$

Combing (31) and (32), we have

$$\lim_{|u_1| \rightarrow \infty} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(u_1, u_2, u_3) = 0. \quad (33)$$

By following similar steps as before we get

$$\lim_{|u_2| \rightarrow \infty} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(u_1, u_2, u_3) = 0, \quad (34)$$

and

$$\lim_{|u_3| \rightarrow \infty} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(u_1, u_2, u_3) = 0. \quad (35)$$

Which completes the proof. $\square$

Next, we will show that the relation between the OCLCT and the 3-D LCT.

Theorem 6. Let an octonion valued function $f : \mathbb{R}^3 \rightarrow \mathbb{O}$ and $L_{A_1, A_2, A_3} \{f\}(\mathbf{u})$ be the 3-D LCT of the function f . Then we have

$$\begin{aligned} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{u}) &= \frac{1}{4} \{ (L_{A_1, A_2, A_3} \{f\}(\mathbf{u}) + L_{A_1, A_2, A'_3} \{f\}(\mathbf{u}))(1 - e_3) \\ &\quad + (L_{A_1, A'_2, A_3} \{f\}(\mathbf{u}) + L_{A_1, A'_2, A'_3} \{f\}(\mathbf{u}))(1 + e_3) \} \\ &\quad + \frac{1}{4} \{ (L_{A_1, A_2, A_3} \{f\}(\mathbf{u}) - L_{A_1, A_2, A'_3} \{f\}(\mathbf{u}))(1 - e_3) \\ &\quad + (L_{A_1, A'_2, A_3} \{f\}(\mathbf{u}) - L_{A_1, A'_2, A'_3} \{f\}(\mathbf{u}))(1 + e_3) \} \cdot e_5. \end{aligned} \quad (36)$$

Proof. The 3-D LCT is defined by

$$L_{A_1, A_2, A_3} \{f\}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{e_1 \gamma_2} e^{e_1 \gamma_3} d\mathbf{x}. \quad (37)$$

$$\text{Let } A'_2 = \begin{bmatrix} a_2 & -b_2 \\ -c_2 & d_2 \end{bmatrix}, \text{ then}$$

$$L_{A_1, A'_2, A_3} \{f\}(\mathbf{u}) = \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{-e_1 \gamma_2} e^{e_1 \gamma_3} d\mathbf{x}. \quad (38)$$

Applying to the equivalent definition of sine and cosine functions, we obtain

$$\begin{aligned} &\frac{1}{2} (L_{A_1, A_2, A_3} \{f\}(\mathbf{u}) + L_{A_1, A'_2, A_3} \{f\}(\mathbf{u})) \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} \cos \gamma_2 e^{e_1 \gamma_3} d\mathbf{x}, \end{aligned} \quad (39)$$

$$\begin{aligned} &\frac{1}{2} (L_{A_1, A'_2, A_3} \{f\}(\mathbf{u}) - L_{A_1, A_2, A_3} \{f\}(\mathbf{u})) \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} (-e_1 \sin \gamma_2) e^{e_1 \gamma_3} d\mathbf{x}. \end{aligned} \quad (40)$$

Multiplying (40) from the right by e_3 and applying the multiplication rules from Table 1, we have

$$\begin{aligned} &\frac{1}{2} (L_{A_1, A'_2, A_3} \{f\}(\mathbf{u}) - L_{A_1, A_2, A_3} \{f\}(\mathbf{u})) e_3 \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} (e_2 \sin \gamma_2) e^{-e_1 \gamma_3} d\mathbf{x}. \end{aligned} \quad (41)$$

Adding (39) and (41), we obtain

$$\begin{aligned} &\frac{1}{2} (L_{A_1, A_2, A_3} \{f\}(\mathbf{u}) + L_{A_1, A'_2, A_3} \{f\}(\mathbf{u})) \\ &\quad + \frac{1}{2} (L_{A_1, A'_2, A_3} \{f\}(\mathbf{u}) - L_{A_1, A_2, A_3} \{f\}(\mathbf{u})) e_3 \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{e_2 \gamma_2} e^{-e_1 \gamma_3} d\mathbf{x}. \end{aligned} \quad (42)$$

To simplify the notation, let

$$\begin{aligned} G_{A_1, A'_2, A_3}^{A_1, A_2, A_3}(\mathbf{u}) &= \frac{1}{2} (L_{A_1, A_2, A_3} \{f\}(\mathbf{u}) + L_{A_1, A'_2, A_3} \{f\}(\mathbf{u})) \\ &\quad + \frac{1}{2} (L_{A_1, A'_2, A_3} \{f\}(\mathbf{u}) - L_{A_1, A_2, A_3} \{f\}(\mathbf{u})) e_3, \end{aligned} \quad (43)$$

$$\text{and } A'_3 = \begin{bmatrix} a_3 & -b_3 \\ -c_3 & d_3 \end{bmatrix}, \text{ then}$$

$$G_{A_1, A'_2, A'_3}^{A_1, A_2, A'_3}(\mathbf{u})$$

$$= \frac{1}{2} (L_{A_1, A_2, A_3'} \{f\}(\mathbf{u}) + L_{A_1, A_2', A_3} \{f\}(\mathbf{u})) + \frac{1}{2} (L_{A_1, A_2', A_3'} \{f\}(\mathbf{u}) - L_{A_1, A_2, A_3'} \{f\}(\mathbf{u})) e_3 \quad (44)$$

$$= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{e_2 \gamma_2} e^{e_1 \gamma_3} d\mathbf{x}. \quad (44)$$

According to the same method, we have

$$\begin{aligned} & \frac{1}{2} (G_{A_1, A_2, A_3}^{A_1, A_2, A_3}(\mathbf{u}) + G_{A_1, A_2', A_3}^{A_1, A_2, A_3'}(\mathbf{u})) \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{e_2 \gamma_2} \cos \gamma_3 d\mathbf{x}, \end{aligned} \quad (45)$$

$$\begin{aligned} & \frac{1}{2} (G_{A_1, A_2, A_3}^{A_1, A_2, A_3}(\mathbf{u}) - G_{A_1, A_2', A_3}^{A_1, A_2, A_3'}(\mathbf{u})) \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{e_2 \gamma_2} (-e_1 \sin \gamma_3) d\mathbf{x}. \end{aligned} \quad (46)$$

Multiplying (46) from the right by e_5 and applying the multiplication rules from Table 1, we have

$$\begin{aligned} & \frac{1}{2} (G_{A_1, A_2, A_3}^{A_1, A_2, A_3}(\mathbf{u}) - G_{A_1, A_2', A_3}^{A_1, A_2, A_3'}(\mathbf{u})) e_5 \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{e_2 \gamma_2} (e_4 \sin \gamma_3) d\mathbf{x}. \end{aligned} \quad (47)$$

Adding (45) and (47), we obtain

$$\begin{aligned} & \frac{1}{2} (G_{A_1, A_2, A_3}^{A_1, A_2, A_3}(\mathbf{u}) + G_{A_1, A_2', A_3}^{A_1, A_2, A_3'}(\mathbf{u})) \\ &+ \frac{1}{2} (G_{A_1, A_2, A_3}^{A_1, A_2, A_3}(\mathbf{u}) - G_{A_1, A_2', A_3}^{A_1, A_2, A_3'}(\mathbf{u})) e_5 \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \int_{\mathbb{R}^3} f(\mathbf{x}) e^{e_1 \gamma_1} e^{e_2 \gamma_2} e^{e_4 \gamma_3} d\mathbf{x} \end{aligned} \quad (48)$$

Finally, substituting (43) and (44) in (48), we obtain the result. $\square$

4. Uncertainty principles of the OCLCT

In signal processing an enormous body of literature exists for study of the uncertainty principle during the last years. In quantum mechanics, an uncertainty principle asserts that one cannot be certain of the position and of the velocity of an electron (or any particle) at the same time. That is, increasing the knowledge of the position decreases the knowledge of the velocity or momentum of an electron. In quaternion analysis, some researches combined the uncertainty relations and the QLCT [13–16,26,31]. In this section we exploit the Heisenberg's uncertainty principle and Donoho-Stark's uncertainty principle for the OCLCT.

Theorem 7. (Heisenberg's uncertainty principle for the OCLCT) Let $f \in L^1(\mathbb{R}^3, \mathbb{O}) \cap L^2(\mathbb{R}^3, \mathbb{O})$, then the following inequality is satisfied:

$$\int_{\mathbb{R}^2} \mathbf{x}^2 |f(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2, A_3}^0 \{f\}(\mathbf{u})|^2 d\mathbf{u} \geq \frac{2}{\pi |b_3|} b_1^2 b_2^2 \|f\|_2^2. \quad (49)$$

Proof. According to (27), we have

$$|f(\mathbf{x})|^2 = |\tilde{f}_e(\mathbf{x})|^2 + |\tilde{f}_o(\mathbf{x})|^2 + |\hat{f}_e(\mathbf{x})|^2 + |\hat{f}_o(\mathbf{x})|^2. \quad (50)$$

Hence

$$\begin{aligned} \int_{\mathbb{R}^2} \mathbf{x}^2 |f(\mathbf{x})|^2 d\mathbf{x} &= \int_{\mathbb{R}^2} \mathbf{x}^2 |\tilde{f}_e(\mathbf{x})|^2 d\mathbf{x} + \int_{\mathbb{R}^2} \mathbf{x}^2 |\tilde{f}_o(\mathbf{x})|^2 d\mathbf{x} \\ &+ \int_{\mathbb{R}^2} \mathbf{x}^2 |\hat{f}_e(\mathbf{x})|^2 d\mathbf{x} + \int_{\mathbb{R}^2} \mathbf{x}^2 |\hat{f}_o(\mathbf{x})|^2 d\mathbf{x}. \end{aligned} \quad (51)$$

From the OCLCT has been divided into four QLCTs in (22), we obtain

$$\begin{aligned} & \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2, A_3}^0 \{f\}(\mathbf{u})|^2 d\mathbf{u} \\ &= \frac{1}{2\pi |b_3|} \left(\int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2}^0 \{\tilde{f}_e\}(\mathbf{u})|^2 d\mathbf{u} + \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2}^0 \{\tilde{f}_o\}(\mathbf{u})|^2 d\mathbf{u} \right. \\ &+ \left. \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2}^0 \{\hat{f}_e\}(\mathbf{u})|^2 d\mathbf{u} + \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2}^0 \{\hat{f}_o\}(\mathbf{u})|^2 d\mathbf{u} \right). \end{aligned} \quad (52)$$

By the Heisenberg's uncertainty principle for the QLCT [30], we get

$$\int_{\mathbb{R}^2} \mathbf{x}^2 |\tilde{f}_e(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2}^0 \{\tilde{f}_e\}(\mathbf{u})|^2 d\mathbf{u} \geq b_1^2 b_2^2 \|\tilde{f}_e\|_2^2. \quad (53)$$

Similar inequalities hold for the remaining three terms. Collecting all, we obtain

$$\int_{\mathbb{R}^2} \mathbf{x}^2 |f(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2, A_3}^0 \{f\}(\mathbf{u})|^2 d\mathbf{u} \geq \frac{2}{\pi |b_3|} b_1^2 b_2^2 \|f\|_2^2. \quad (54)$$

Which completes the proof. $\square$

Remark 2. If $\|f\|_2 = 1$, then we have

$$\int_{\mathbb{R}^2} \mathbf{x}^2 |f(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^2} \mathbf{u}^2 |\mathcal{L}_{A_1, A_2, A_3}^0 \{f\}(\mathbf{u})|^2 d\mathbf{u} \geq \frac{2}{\pi |b_3|} b_1^2 b_2^2.$$

Theorem 8. (Donoho-Stark's uncertainty principle of the OCLCT) Let Ω and Λ be two measurable subsets of $\mathbb{R}^3$ and $f \in L^1(\mathbb{R}^3, \mathbb{O}) \cap L^2(\mathbb{R}^3, \mathbb{O})$. If f is ε_Ω -concentration to Ω in $L^1(\mathbb{R}^3, \mathbb{O})$ -norm and $\mathcal{L}_{A_1, A_2, A_3}^0 \{f\}$ is ε_Λ -concentration to Λ in $L^2(\mathbb{R}^3, \mathbb{O})$ -norm, then

$$(1 - \varepsilon_\Omega)(1 - \varepsilon_\Lambda) \leq \frac{|b_1 b_2|^{-\frac{1}{2}}}{2\pi} \sqrt{|\Omega||\Lambda|}. \quad (55)$$

Proof. In [29], the authors proved the Donoho-Stark's uncertainty principle-type of the QLCT as follows

$$\|\mathcal{L}_{A_1, A_2} \{f\}\|_2^2 \leq \frac{|b_1 b_2|^{-1} |\Omega||\Lambda|}{[2\pi(1 - \varepsilon_\Omega)(1 - \varepsilon_\Lambda)]^2} \|f\|_2^2. \quad (56)$$

According to (25), we know that the OCLCT is a combination of four QLCT. Hence

$$\begin{aligned} \|\mathcal{L}_{A_1, A_2}^0 \{f\}\|_2^2 &\leq \frac{|\Omega||\Lambda|}{2\pi |b_1 b_2 b_3| [2\pi(1 - \varepsilon_\Omega)(1 - \varepsilon_\Lambda)]^2} \|\tilde{f}_e\|_2^2 + \|\tilde{f}_o\|_2^2 + \|\hat{f}_e\|_2^2 + \|\hat{f}_o\|_2^2 \\ &= \frac{|\Omega||\Lambda|}{8\pi^3 |b_1 b_2 b_3| [(1 - \varepsilon_\Omega)(1 - \varepsilon_\Lambda)]^2} \|f\|_2^2, \end{aligned} \quad (57)$$

where the last equality may be obtained by (27).

From Theorem 4, we obtain

$$(1 - \varepsilon_\Omega)(1 - \varepsilon_\Lambda) \leq \frac{|b_1 b_2|^{-\frac{1}{2}}}{2\pi} \sqrt{|\Omega||\Lambda|}.$$

Which completes the proof. $\square$

5. The potential applications

In this section, we will present the potential applications to show the importance of the OCLCT. The OCLCT transforms a octonion 3-D signal into a octonion-valued frequency domain signal, which is an effective processing tool for analysing the 3-D linear time-invariant systems of linear partial differential equations with constant coefficients [21].

The image watermarking has been developed in the last decade [25,32,33]. The OCLCT can be applied to analyse the various watermarking techniques based on embedding a color watermark into a chosen part of the octonion hypercomplex spectrum. The idea is that the watermark can be embedded in the octonion linear canonical spectrum coefficients of the original image according to the chosen watermark embedding algorithm. In addition, based on the comparison of OCLCT to the original and watermarked images, and as a result, the extraction of a watermark. The general OCLCT-image watermarking detection scheme is proposed.

The potential application can also be found in recovery of bandlimited octonion 3-D signals in OCLCT domain from noisy samples. It is based on the oversampling theorem and without adding too much complexity, a reconstruction algorithm for bandlimited signals in OCLCT domain from noisy observations is obtained [34].

As a theoretical basis of the frequency-domain definitions of multidimensional analytic signals [25], the OCLCT can be applied in the domain of analytic signals. Other potential applications can be found in physics, filter design, color image analysis and processing, such as the methods presented in [35–38].

The Heisenberg's uncertainty principle for the OCLCT states that the relation of one octonion-valued signal in spatial and another octonion-valued signal in frequency domain. A potential application can be found in the estimation of the lower bound of integral [39–41]. Such as if a octonion 3-D signal in Theorem 7 is determined, it is difficult to obtain the value of the left side of Theorem 7 by calculating the integral. However, an estimation of the left side of Theorem 7 can be easily obtained based on Theorem 7.

On the other hand, the Heisenberg's uncertainty principle implies that it defines a lower bound for the product of a signal spread and its bandwidth. So, we can also estimate the bandwidth by the uncertainty principle [41,42]. For example, if T_A is known, then the bandwidth of a system that performs an OCLCT cannot be narrower than $\frac{2}{T_A^2 \pi |b_3|} b_1^2 b_2^2$.

Moreover, Donoho-Stark's uncertainty principle plays an important role in signal reconstruction from a noisy measurement [31,43–45]. Then, the Donoho-Stark's uncertainty principle for the OCLCT can be applied in 3-D hypercomplex analytic signals reconstruction from a noisy measurement.

6. Conclusions

In the present work, we present the Cayley-Dickson hypercomplex LCT-the OCLCT. We get the relationship between the OCLCT and the QLCT. Based on the relation, the various properties of the OCLCT are explored, such as shift property, inversion formula, isometry and Riemann-Lebesgue lemma. The relation between OCLCT and 3-D LCT is also obtained. Furthermore, the Heisenberg's uncertainty principle and Donoho-Stark's uncertainty principle associated with the OCLCT are obtained. Finally, some potential applications of the OCLCT are proposed. Further investigations on this topic are now under investigation such as convolution theorem and sampling theorem. They will be reported in the following paper. The real applications of the OCLCT are also the future research directions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Wen-Biao Gao: Conceptualization, Methodology, Validation, Formal analysis, Writing - original draft. **Bing-Zhao Li:** Validation, Resources, Supervision, Project administration, Funding acquisition.

Acknowledgement

This work is supported by the National Natural Science Foundation of China (No. 61671063).

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.sigpro.2021.108233](https://doi.org/10.1016/j.sigpro.2021.108233).

References

- [1] K.M. Snopek, The study of properties of n-d analytic signals and their spectra in complex and hypercomplex domains, *Radio Eng.* 21 (1) (2012) 29–36.
- [2] S.C. Pei, J.H. Chang, J.J. Ding, Color pattern recognition by quaternion correlation, in: *IEEE International Conference Image Process.*, Thessaloniki, Greece, October 7–10, 2010, pp. 894–897.
- [3] S.J. Sangwine, T.A. Ell, Colour image filters based on hypercomplex convolution, *IEEE Proc. Vis. Image Signal Process.* 49 (21) (2000) 89–93.
- [4] S.J. Sangwine, C.J. Evans, T.A. Ell, Colour-sensitive edge detection using hypercomplex filters, in: *Proceedings of the 10th European Signal Processing Conference EUSIPCO*, Tampere, Finland, 1, 2000, pp. 107–110.
- [5] B. Witten, J. Shragge, Quaternion-based signal processing, *stanford exploration project*, New Orleans Annu. Meet. (2006) 2862–2866.
- [6] C. Gao, J. Zhou, F. Lang, Q. Pu, C. Liu, Novel approach to edge detection of color image based on quaternion fractional directional differentiation, *Adv. Autom. Robot.* 1 (2012) 163–170.
- [7] C.C. Took, D.P. Mandic, The quaternion LMS algorithm for adaptive filtering of hypercomplex processes, *IEEE Trans. Signal Process.* 57 (4) (2009) 1316–1327.
- [8] T. Bülöw, G. Sommer, The hypercomplex signal-a novel extensions of the analytic signal to the multidimensional case, *IEEE Trans. Signal Process.* 49 (11) (2001) 2844–2852.
- [9] H. De Bie, *Fourier transforms in clifford analysis*, *Operator Theory* edited by Daniel Alpay, Springer Basel, 2015.
- [10] S.J. Sangwine, T.A. Ell, Hypercomplex Fourier transforms of color images, *IEEE Trans. Image Process.* 16 (1) (2007) 22–35.
- [11] E. Bayro-Corrochano, N. Trujillo, M. Naranjo, Quaternion Fourier descriptors for preprocessing and recognition of spoken words using images of spatiotemporal representations, *J. Math. Imaging Vis.* 28 (2) (2007) 179–190.
- [12] P. Bas, N. LeBihan, J.M. Chassery, Color image water marking using quaternion Fourier transform, in: *Proceedings of the IEEE International Conference on Acoustics Speech and Signal Processing*, ICASSP, HongKong, 2003, pp. 521–524.
- [13] K.I. Kou, J. Ou, J. Morais, Uncertainty principles associated with quaternionic linear canonical transforms, *Math. Meth. Appl. Sci.* 39 (10) (2016) 2722–2736.
- [14] M.E. Kassimi, Y.E. Haoui, S. Fahlaoui, The Wigner-Ville distribution associated with the quaternion offset linear canonical transform, *Anal. Math.* 45 (4) (2019) 787–802.
- [15] Y. Yang, K.I. Kou, On uncertainty principles for hypercomplex signals in the linear canonical transform domains, *Signal Process.* 95 (2014) 67–75.
- [16] Y.N. Zhang, B.Z. Li, Novel uncertainty principles for two-sided quaternion linear canonical transform, *Adv. Appl. Clifford Algebr.* 28 (1) (2018) 1–14.
- [17] J. Kauhanen, H. Orelma, Cauchy-Riemann operators in octonionic analysis, *Adv. Appl. Clifford Algebr.* 28 (1) (2018) 1–14.
- [18] S. Hahn, K. Snopek, The unified theory of n-dimensional complex and hypercomplex analytic signals, *Bull. Polish Ac. Sci. Tech. Sci.* 59 (2011) 167–181.
- [19] L. Błaszczczyk, K. Snopek, Octonion Fourier transform of real-valued functions of three variables-selected properties and examples, *Signal Process.* 136 (2017) 29–37.
- [20] P. Lian, The octonionic Fourier transform: uncertainty relations and convolution, *Signal Process.* 164 (2019) 295–300.
- [21] L. Błaszczczyk, A generalization of the octonion Fourier transform to 3-D octonion-valued signals: properties and possible applications to 3-D LTI partial differential systems, *Multidimens. Syst. Signal Process.* 31 (4) (2020) 1227–1257.

- [22] L. Blaszczyk, Discrete octonion Fourier transform and the analysis of discrete 3-D data, *Comput. Appl. Math.* 39 (4) (2020) 1–19.
- [23] J.H. Conway, D.A. Smith, *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*, A K Peters Ltd, Natick, 2003.
- [24] K. Snopek, New hypercomplex analytic signals and Fourier transforms in Cayley-Dickson algebras, *Electron. Telecommun. Q.* 55 (3) (2009) 403–415.
- [25] S.L. Hahn, K.M. Snopek, *Complex and Hypercomplex Analytic Signals: Theory and Applications*, Artech House, Norwood, 2016.
- [26] W.B. Gao, B.Z. Li, Quaternion windowed linear canonical transform of two-dimensional quaternionic signals, *Adv. Appl. Clifford Algebr.* 30 (1) (2020) 1–18.
- [27] K.I. Kou, J. Ou, J. Morais, On uncertainty principle for quaternionic linear canonical transform, *Abstr. Appl. Anal.* (2013) 1–14.
- [28] K.I. Kou, J. Morais, Asymptotic behaviour of the quaternion linear canonical transform and the Bochner-Minlos theorem, *Appl. Math. Comput.* 247 (15) (2014) 675–688.
- [29] A. Achak, A. Abouelaz, R. Daher, N. Safouane, Uncertainty principles for the quaternion linear canonical transform, *Adv. Appl. Clifford Algebr.* 29 (5) (2019) 1–19.
- [30] B. Mawardi, R. Ashino, A simplified proof of uncertainty principle for quaternion linear canonical transform, *Abstr. Appl. Anal.* (2016) 1–11.
- [31] X.Y. Zhu, S.Z. Zheng, Uncertainty principles for the two-sided quaternion linear canonical transform, *Circuits Syst. Signal Process.* 39 (9) (2020) 4436–4458.
- [32] X. Wang, C. Wang, H. Yang, P. Niu, A robust blind color image watermarking in quaternion Fourier transform domain, *J. Syst. Softw.* 86 (2) (2013) 255–277.
- [33] C. Li, B. Li, L. Xiao, Y. Hu, L. Tian, A watermarking method based on hypercomplex Fourier transform and visual attention, *J. Inf. Comput. Sci.* 9 (15) (2012) 4485–4492.
- [34] H. Zhao, R.Y. Wang, D.P. Song, Recovery of bandlimited signals in linear canonical transform domain from noisy samples, *Circuits Syst. Signal Process.* 33 (6) (2014) 1997–2008.
- [35] P.R. Girard, *Quaternions, Clifford Algebras and Relativistic Physics*, Birkhuser, Boston, MA, 2008.
- [36] M. Bahri, Quaternion linear canonical transform application, *Glob. J. Pure Appl. Math.* 11 (1) (2005) 19–24.
- [37] D.S. Alexiadis, G.D. Sergiadis, Estimation of motions in color image sequences using hypercomplex Fourier transforms, *IEEE Trans. Image Process.* 18 (1) (2009) 168–187.
- [38] T. Ell, S. Sangwine, Decomposition of 2D hypercomplex Fourier transforms into Pairs of Fourier transforms, in: *Proc. EUSIPCO*, Vol. II, Tampere, Finland, 2000.
- [39] P. Dang, G.T. Deng, T. Qian, A tighter uncertainty principle for linear canonical transform in terms of phase derivative, *IEEE Trans. Signal Process.* 61 (21) (2013) 5153–5164.
- [40] Z.C. Zhang, Uncertainty principle for linear canonical transform using matrix decomposition of absolute spread matrix, *Digit. Signal Process.* 89 (2019) 145–154.
- [41] W.B. Gao, B.Z. Li, Uncertainty principles for the short-time linear canonical transform of complex signals, *Digit. Signal Process.* 111 (2021) 1–10.
- [42] A. Stern, Uncertainty principles in linear canonical transform domains and some of their implications in optics, *J. Opt. Soc. Am. A-Opt. Image Sci. Vis.* 25 (3) (2008) 647–652.
- [43] A.H. Jeffrey, D.L. Joseph, *Time-frequency and time-scale methods: adaptive decompositions, Uncertainty Principles, and Sampling*, Birkhuser(2004).
- [44] M.S. Liu, K.I. Kou, J. Morais, P. Dang, Sharper uncertainty principles for the windowed Fourier transform, *J. Mod. Opt.* 62 (2015) 46–55.
- [45] D.L. Donoho, P.B. Stark, Uncertainty principles and signal recovery, *SIAM J. Appl. Math.* 49 (3) (1989) 906–931.